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Silviculture and planted species interact to influence reforestation success on a *Kalmia*-dominated site – a 15-year study

by Nelson Thiffault¹, Brian D. Titus² and Martin T. Moroni³

ABSTRACT

Successful regeneration following harvesting or natural disturbance is a fundamental prerequisite for sustainable forest management. However, some regenerating stands have poor juvenile growth rates, which compromise sustainable management objectives. In particular, the presence of some ericaceous species that proliferate after forest disturbance, such as *Kalmia angustifolia*, can slow succession of boreal stands to the point that ecosystem retrogression is induced. We used data from a silvicultural field trial established in central Newfoundland to evaluate how various combinations of silvicultural treatments (trench scarification, herbicide application, fertilization at planting) influenced growth of three conifer species planted on a *Kalmia*-dominated cutover. Ground-level diameter (GLD), height, diameter at breast height (DBH), and percent *Kalmia* cover were assessed at the end of 15 growing seasons after planting. We detected several interactions between silvicultural treatments and planted conifer species. Globally, height and estimates of foliar biomass of all conifer species responded positively to scarification. Fifteen-year height in both scarified and unscarified treatments was in the order *Picea mariana* < *Pinus banksiana* < *Larix laricina*. Black spruce and jack pine height increased when *Kalmia* was controlled with herbicide, but height of tamarack was not. The use of herbicide significantly increased 15-year GLD and volume index of all three conifer species, but only black spruce responded positively to fertilization at planting. Our results confirm that species-specific responses to silvicultural treatments are to be expected when managing *Kalmia*-dominated sites. Although chemical vegetation management has great silvicultural potential, our results suggest that mechanical site preparation can also be effective in promoting early conifer seedling growth that leads to rapid canopy closure. It is anticipated that canopy closure will lead to exclusion of *Kalmia* later in the rotation through natural successional pathways.

Key words: *Kalmia angustifolia*, vegetation management, scarification, herbicide, fertilizer, black spruce, jack pine, tamarack

RÉSUMÉ

L'établissement réussi de la régénération forestière après les perturbations naturelles ou la coupe forestière est un élément critique de l'aménagement forestier durable (AFD). Toutefois, certains peuplements présentent une faible croissance juvénile, ce qui compromet l'atteinte des objectifs d'AFD. En particulier, la présence d'espèces éricacées qui prolifèrent avec les perturbations, telles le *Kalmia angustifolia*, peut ralentir la succession forestière de peuplements boréaux au point d'induire une rétrogression de l'écosystème. Nous avons utilisé les données issues d'un dispositif sylvicole établi au centre de Terre-Neuve (Canada), pour évaluer comment différentes combinaisons de traitements sylvicoles (scarifiage à disques, application d'un herbicide, fertilisation au moment de la mise en terre) influencent la croissance de trois espèces de conifères plantées sur une station envahie par le *Kalmia*. Le diamètre au niveau du sol (DHS), la hauteur, le diamètre à hauteur de poitrine et la couverture en *Kalmia* ont été évalués 15 saisons de croissance après la mise en terre des plants. Nous avons détecté plusieurs interactions entre les traitements sylvicoles et l'espèce plantée. Globalement, la hauteur et la biomasse foliaire estimée de toutes les espèces ont répondu positivement au scarifiage à disques. La hauteur à 15 ans était dans l'ordre *Picea mariana* < *Pinus banksiana* < *Larix laricina*, et ce, tant dans les parcelles scarifiées que non scarifiées. La hauteur de l'épinette noire et du pin gris a augmenté lorsque le *Kalmia* a été contrôlé avec un herbicide, alors que la hauteur du mélèze n'y a pas réagi. L'utilisation d'herbicide a significativement augmenté le DHS à 15 ans et l'index de volume des trois espèces, mais seule l'épinette noire a répondu positivement au traitement de fertilisation. Nos résultats confirment que des réponses spécifiques aux traitements sylvicoles sont à prévoir dans l'aménagement des stations dominées par le *Kalmia*. Bien que la gestion chimique de la végétation présente un grand potentiel sylvicole, nos résultats suggèrent que la préparation mécanique du sol, un traitement considéré plus compatible avec l'aménagement écosystémique, peut également être efficace et favoriser la croissance initiale des plants, ce qui mènera à une fermeture rapide du couvert. Nous anticipons que la fermeture du couvert résultera en l'exclusion graduelle du *Kalmia* plus tard dans la révolution, suivant les trajectoires successioneuses naturelles.

Mots-clés : *Kalmia angustifolia*, gestion de la végétation, scarifiage, herbicide, engrais, épinette noire, pin gris, mélèze laricin

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Introduction

Some regenerating stands have poor juvenile growth rates which can compromise sustainable forest management objectives. In particular, the presence of some ericaceous species can slow forest succession to a point where ecosystem retrogression is induced (Nilsson and Wardle 2005). Such shifts from once-productive forest stands to ericaceous-dominated ecosystems have been documented in

various ecosystems in Europe (e.g., *Vaccinium myrtillus* L., Mallik and Pellissier 2000; *Calluna vulgaris* L. (Hull), Read and Jalal 1980; *Empetrum hermaphroditum* Hagerup, Zackrisson *et al.* 1997) and North America (e.g., *Rhododendron maximum* L., Nilsen *et al.* 1999; *Gaultheria shallon* Pursh, Prescott and Sajedi 2008). On some sites in eastern Canada, rapid encroachment and site domination by *Kalmia angustifolia* L. constitutes another well-documented example of this phenomenon (Mallik 2003).

A combination of mechanisms is thought to be responsible for the effects of *Kalmia* and related species on conifer establishment and growth. The allelopathic properties of phenolic compounds found in *Kalmia* leaves have been demonstrated in laboratory and greenhouse studies (e.g., Mallik 1987, Zhu and Mallik 1994). The impact of direct competition for nutrients by *Kalmia* and *Vaccinium* spp. on conifer regeneration success has been shown in the field using stable isotopes (Thiffault *et al.* 2004b). More recently, it has been confirmed that *Kalmia* litter increases the amount of soil N sequestered as tannin-protein complexes, which may improve the competitive ability of the species relative to black spruce (*Picea mariana* (Mill.) BSP) (Joanisse *et al.* 2009). Furthermore, the various nutrient cycling processes in spruce-*Kalmia* associations interact and can vary, depending on site quality (Yamasaki *et al.* 2002). Regardless of the specific mechanisms involved, direct competition for nutrients and indirect nutritional interference appear to be the key factors explaining the chlorotic appearance and stunted growth of conifers regenerating on some *Kalmia*-dominated sites.

Kalmia and associated species can be controlled effectively with chemical herbicides, and a >95% decrease in *Kalmia*

cover was still found three years following herbicide application on two sites heavily dominated by *Kalmia* in Newfoundland (Canada) (English and Titus 2000) and control continued to be effective after five years (Titus and English 2000). Elsewhere, reduction of *Kalmia* cover can improve the early growth and nutrition of planted seedlings, resulting in fundamental changes in the biochemical properties of the forest floor (LeBel *et al.* 2008).

Although chemical vegetation management has great silvicultural potential for releasing plantations (Dampier *et al.* 2006) and has limited negative impacts on biotic elements of the ecosystem (Lautenschlager and Sullivan 2002), public acceptance of herbicides is low (Wagner *et al.* 1998). Moreover, worldwide certification systems discourage the use of chemical herbicides (e.g., Forest Stewardship Council 1996). Many authors have recommended intense mechanical site preparation for managing heaths (e.g., Zehetmayr 1960), and more specifically, *Kalmia*-dominated sites (e.g., Richardson 1981, Mallik 1994, Titus *et al.* 1995, Yamasaki *et al.* 2002). Silvicultural trials conducted in northern Quebec (Canada) have confirmed the short-term impact of soil scarification on planted black spruce and jack pine (*Pinus banksiana* Lamb.) seedling growth for sites dominated by *Kalmia* and related species (Thiffault *et al.* 2005, Thiffault and Jobidon 2006). Despite these promising results, longer-term assessments of scarification effects on tree growth on *Kalmia*-dominated sites are scarce. Ten-year follow-up of an experiment established in northeastern Quebec suggested that scarification allowed a better expression of site index, implying a long-term impact on microsite characteristics (Thiffault *et al.* 2004a). However, this study was limited in scope, as it comprised only two replicates, and the size of the experimental units was too small to warrant longer-term assessment of tree performance.

Given the important nutritional component of *Kalmia* interference, fertilization also offers great potential for alleviating the competitive pressure that this species can induce (cf. Taylor and Tabbush 1990). Recent studies conclude that there can be a significant but short-lived beneficial effect on early growth of planted conifer seedlings (Thiffault *et al.* 2004a, Thiffault and Jobidon 2006, LeBel *et al.* 2008). Longer-term beneficial effects on growth can be expected, however, as increased early seedling performance could favour accelerated canopy closure that could in turn influence nutritional site quality, as observed in another ericad-conifer ecosystem in coastal B.C. (Bradley *et al.* 2000). Indeed, recent field sampling on the site used in the current study confirmed that fertility of the soil organic layer increased when conifer dominance was re-established, even in the presence of significant *Kalmia* re-growth (Moroni *et al.* 2009).

Our objectives in the current study were to evaluate how various combinations of silvicultural treatments and planted species influenced 15-year total growth of conifers on a *Kalmia*-dominated site in central Newfoundland. We posited that (i) scarification would lead to bigger trees, (ii) herbicide application would have a long-term impact on *Kalmia* cover and tree growth, (iii) the effects of fertilization at the time of planting on seedling growth would still be detectable 15 years after planting, and (iv) *Kalmia* would affect planted jack pine and tamarack (*Larix laricina* [Du Roi] K. Koch) less than black spruce because of their relative differences in nutritional requirements, resulting in species-specific responses to silvi-

cultural treatments. To our knowledge, this study is the most complex silvicultural trial to date on the regeneration of *Kalmia*-dominated sites, and covers the broad scope of reasonable silvicultural options that could be used on these sites in the same trial. We thus believe that it provides the strongest evidence for management recommendations for similar sites in northeastern North America.

Materials and Methods

Site description

An experimental plantation was established in central Newfoundland, near Botwood (lat. 49°07'60"N; long. 55°26'16"W). The climate is humid continental with cool summers and no dry season, with a mean annual temperature of 4.3°C, 185 days per year with a minimum temperature >0°C, and an average annual precipitation (1971–2000) of 986 mm (of which 24% fell as snow). The uniform, level site is located on a glaciofluvial terrace deposit. The Orthic Humo–Ferric Podzol (Soil Classification Working Group 1998) is covered by a mor humus layer 9 cm to 12 cm thick, and characterized by Ae, Bf and Bfh horizons of 3 cm to 8 cm, 9 cm to 10 cm, and 10 cm to 16 cm thick, respectively. The texture of the A and B horizons ranges from silt loam to very fine sandy loam. The Damman forest type is classified as *Pleurozium* Balsam Fir, with some areas transitional to *Hylocomium* Balsam Fir (Meades and Moores 1994). The previous stand was dominated by black spruce, was commercially thinned in 1973–1974 at age 65 years to 70 years, was burned in a wildfire in 1979, and salvage cut in 1982, all of which stimulated the establishment and spread of *Kalmia*. Although *Kalmia* was dominant, other species were present, such as *Ahnus crispus* (Ait.) Pursh, *Amelanchier bartramiana* (Taush) Roemer, *Betula papyrifera* Marsh., *Prunus pennsylvanica* L.f., *Salix* spp., *Viburnum cassinoides* L., and *Vaccinium* spp.

Experimental design and treatments

The field trial was established in 1989 and consisted of 36 treatment units (2 scarification levels × 6 factorial herbicide+fertilizer combinations × 3 conifer species) combined in a complete block, split-split-plot design with four replicates, resulting in a total of 144 experimental units. Each experimental block (replicate) covered 1.35 ha and was split in half, one of which was randomly selected for scarification (S1) while the other one was left unscarified (S0). Scarification was performed in the selected main plots in the fall of 1989 with a TTS disk trencher. Each main plot was divided into six 30 × 30 m subplots with a 5-m buffer strip between them. Only four subplots are considered in the present study, however, as one of the three fertilizers treatments (NPK + micronutrients) contained a then-new micronutrient formulation designed to counteract soil acidity (FYT0, Biofutura Oy Ltd., Espoo, Finland) that is now considered to be operationally unrealistic in Newfoundland. Every subplot was randomly assigned to receive one of four factorial combinations of herbicide (H) + fertilizer (F) treatments (H0F0 = no herbicide, no fertilization; H0F1 = no herbicide, with fertilization; H1F0 = with herbicide, no fertilization; H1F1 = with herbicide, with fertilization). This study pre-dated later work that determined effective chemical control methods (English and Titus 2000, Titus and English 2000) and was not designed to be a herbicide efficacy trial. Based on the best available expert opinion at the time, the herbicide applications thus consisted of a combination of

treatments: hexazinone (Velpar L[®]) at 18 L (4.32 kg a.i.) ha⁻¹ in 300 L ha⁻¹ water and applied pre-bud break in late May 1989, triclopyr (Garlon 4[®]) at 8 L ha⁻¹ in 300 L ha⁻¹ water (3.84 kg a.i. ha⁻¹) and applied just after full leaf extension in mid-July 1989, and spot application of glyphosate (Vision[®]) at 6 L ha⁻¹ in 200 L ha⁻¹ water (2.88 kg a.i. ha⁻¹) carried out in mid-September 1989, and late June 1992 on ericaceous plants that survived earlier treatments (planted conifers were covered with plastic bags for protection). Fertilizer was hand-broadcast at the time of planting to emulate aerial application, based on U.K. operations and formulations common at the time (Taylor and Tabbush 1990): 15-15-15 NPK as urea, triple super phosphate and muriate of potash at a rate of 150, 138 and 120 kg ha⁻¹ of N, P and K, respectively.

Subplots were further split into three 10 m × 30 m sub-subplots, each of which was planted in June–July 1990 with either black spruce (Carmenville, NL seed), jack pine (Chibougama, QC seed), or tamarack (Saguenay, 88098C seed) containerized stock (8.5-cm-deep × 3.3-cm-diameter cavities) grown over the previous winter in greenhouses. Each sub-subplot contained 150 seedlings, planted in 5 rows of 30 seedlings each. Seedlings in scarified plots were planted in the hinge position (the trench–berm interface). Half of the seedlings were thinned out over the first five years, leaving a maximum of 75 seedlings at 2 m × 2 m spacing. Each experimental unit contained between 23 and 50 living saplings at the time of measurement after 15 growing seasons.

Saplings and *Kalmia* measurements

Ground-level diameter, total height, current year's leader length, and diameter at breast height (DBH; 130 cm) of saplings were assessed in August–September 2004, 15 full growing seasons after planting. At the same time, percent *Kalmia* cover in a 0.5-m radius plot centred on individual living trees was visually estimated and assigned to one of five cover classes (0%, <25%, 25%–50%, 50%–75% and 75%–100%).

Data analyses

For saplings with height <130 cm, an individual stem volume index was estimated using the volume of a cone:

$$[1] \quad V = \frac{\pi(GLD)^2 H}{12000}$$

where V = stem volume index, in dm³
 GLD = ground-level diameter, in cm
 H = height, in cm

For saplings >130 cm in height, the volume of a truncated cone from ground up to 130 cm was added to the volume of a cone from 130 cm upwards:

$$[2] \quad V = \frac{13\pi}{1200} \times (GLD^2 + GLD \times DBH + DBH^2) + \frac{\pi(DBH)^2(H - 130)}{12000}$$

where V = stem volume index, in dm³
 GLD = ground-level diameter, in cm
 DBH = diameter at breast height (130 cm), in cm
 H = height, in cm

Analyses of variance for mixed models (ANOVA) using the MIXED procedure of SAS (v.9, SAS Institute Inc., Cary, NC) were used to test for significant treatment effects. Scarification (2 levels) was considered as the main plot effect, herbicide+fertilizer (4 levels) as the subplot effect, and conifer species (3 levels) were considered at the sub-subplot level. We tested data for ANOVA prerequisites using standard graphical methods to identify data that were not normally distributed or that had heterogeneous variances and that therefore required transformation before analyses. Total height, 15th-year leader growth, GLD, height:GLD ratio, and volume data were natural-log transformed and we present back-transformed means and 95% confidence intervals with bias correction for these variables (Ung and Végiard 1988, Végiard and Ung 1993). Individual tree foliar biomass was estimated using DBH and height based on Lambert *et al.* (2005):

$$[3] \quad y_{\text{foliage}} = \beta_{\text{foliage1}} \times DBH^{\beta_{\text{foliage2}}} \times H^{\beta_{\text{foliage3}}}$$

where y_{foliage} = estimated foliage biomass (including twigs), in kg

β_{foliage1} = 0.2048, 0.0325, and 0.1324 for black spruce, jack pine and tamarack, respectively

β_{foliage2} = 2.5754, 1.7879, and 2.1140 for black spruce, jack pine and tamarack, respectively

β_{foliage3} = -1.3704, 0, and -0.8781 for black spruce, jack pine and tamarack, respectively

The accuracy of these estimates varies according to species, with adjusted coefficient of determinations ranging from 0.52 for tamarack to 0.70 for jack pine (Lambert *et al.* 2005). Foliar biomass data were square-root transformed before analyses; back-transformed means and 95% confidence intervals with bias correction are presented. Treatment effects on percent *Kalmia* cover classes (an ordinal, multinomial variable) were evaluated using a generalized linear mixed model with the GLIMMIX procedure of SAS. We specified a *cumlogit* link function, and modeled the probabilities of *Kalmia* cover classes being low. Afterwards, *Kalmia* cover

was included as a categorical predictive variable in the model to investigate its sole and interaction effects on sapling leader growth. This analysis was restricted to unherbicide treatments, as *Kalmia* cover classes in herbicide treatments were mostly limited to the 0% and <25% cover classes.

We used Fisher's protected LSD test (Steel *et al.* 1997) in all analyses to separate treatment means when ANOVAs showed significant effects (at $\alpha = 0.05$). Further comparisons were performed with orthogonal contrasts when needed using Bonferroni corrections of the statistical significance level.

Results

Total height and 15th-year leader growth

The herbicide+fertilization effects on 15-year height varied by scarification treatment (significant $S \times HF$ interaction; Table 1). Saplings outplanted in herbicide treatments were significantly taller than those growing in the corresponding control treatments (Table 2), but the herbicide effect was reduced when the soil was scarified (Fig. 1A). Fertilization had no significant effect on height, either in scarified or unscarified treatments (Table 2). Species differences in height were influenced by scarification and herbicide+fertilization (significant $S \times SP$ and $HF \times SP$ interactions; Table 1). All species responded positively to the scarification treatment, and 15-year total height in both scarified and unscarified treatments was in the order spruce < pine < tamarack (Table 2, Fig. 1B). When planted in scarified treatments, jack pine attained a similar height to tamarack planted in unscarified treatments ($p = 0.137$). Black spruce and jack pine total height was significantly increased by control of *Kalmia* with herbicide, but height of tamarack was not (Table 2; Fig. 1C). Fertilization increased black spruce 15-year height by 30% but had no effect on the other two species (Table 2).

Fifteenth-year leader growth was in the order spruce < pine < tamarack in both scarified and unscarified treatments (Table 2), and ranged from 28 cm (black spruce) to 46 cm (tamarack) in the scarified treatment, while it was 16 cm for black spruce, 37 cm for jack pine, and 50 cm for tamarack in the unscarified treatment. However, only black spruce leader growth responded significantly to scarification ($p < 0.001$),

Table 1. ANOVA results (*p*-values) for 15-y dimensions of black spruce (*Picea mariana*), jack pine (*Pinus banksiana*), and tamarack (*Larix laricina*) saplings planted on a *Kalmia*-dominated site in central Newfoundland with various combinations of silvicultural treatments.

Source of variation (fixed)	dfn	dfd	F-value	Height	15 th -year leader growth	GLD	Volume index	Estimated foliage biomass
Scarification (S)	1	19.8	63.59	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Herbicide-Fertilization (HF)	3	19.8	37.45	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
$S \times HF$	3	19.8	5.82	0.005	0.138	0.013	0.006	0.311
Species (SP)	2	45.7	473.21	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
$S \times SP$	2	45.7	13.47	< 0.001	< 0.001	< 0.001	< 0.001	0.351
$HF \times SP$	6	45.7	12.19	< 0.001	< 0.001	< 0.001	< 0.001	0.017
$S \times HF \times SP$	6	45.7	1.53	0.191	0.245	0.232	0.130	0.528

Note: GLD = ground-level diameter; dfn = numerator degrees of freedom for all variables; dfd = denominator degrees of freedom for height, calculated using Satterthwaite's formula (Littell *et al.* 1996). dfd vary from 65.2 to 65.5 for 15th-year leader growth, from 19.9 to 45.5 for GLD, from 19.8 to 45.4 for volume, and from 23.0 to 55.3 for foliage biomass. F-values are for height. F vary from 1.36 to 188.87 for 15th-year leader growth, from 1.41 to 240.81 for GLD, from 1.73 to 336.06 for volume, and from 0.86 to 135.72 for foliage biomass. Analyses of height, GLD, and volume were performed on natural-log transformed data; analyse of foliage biomass was performed on square-root transformed data. Estimates of foliage biomass are based on Lambert *et al.* (2005). Given the significant interactions, values in bold typeface are for interpretation.

Table 2. Orthogonal contrasts for selected variables describing 15-y-old black spruce (*Picea mariana*), jack pine (*Pinus banksiana*), and tamarack (*Larix laricina*) saplings planted on a *Kalmia*-dominated site in central Newfoundland that received various combinations of silvicultural treatments.

Contrast	Total height		15 th -y leader growth		Ground-level diameter		Estimated foliage biomass	
	df	p-value	df	p-value	df	p-value	df	p-value
Scarification × Herbicide-Fertilization								
With herbicide vs. No herbicide								
Non scarified	19.7	<0.001	–	–	19.8	<0.001	–	–
Scarified	19.9	<0.001	–	–	20	<0.001	–	–
With fertilizer vs. No fertilizer								
Non scarified	19.7	0.018	–	–	19.8	0.012	–	–
Scarified	19.9	0.072	–	–	20	0.082	–	–
Scarification × Species								
Eastern larch vs. Jack pine								
Non scarified	45.7	<0.001	65.2	<0.001	45.5	<0.001	–	–
Scarified	45.6	<0.001	65.2	0.004	45.4	0.009	–	–
Herbicide-Fertilization × Species								
With herbicide vs. No herbicide								
Black spruce	53.5	<0.001	65.4	<0.001	51.6	<0.001	138	<0.001
Jack pine	53.5	<0.001	65.3	0.062	51.5	<0.001	56.7	<0.001
Tamarack	53.5	0.008	65.5	0.841	51.6	<0.001	55.1	<0.001
With fertilizer vs. No fertilizer								
Black spruce	53.5	<0.001	65.4	0.005	51.6	0.002	139	<0.001
Jack pine	53.5	0.103	65.3	0.856	51.5	0.03	56.5	0.014
Tamarack	53.5	0.167	65.4	0.255	51.6	0.091	55	0.006

Note: df = degrees of freedom calculated using Satterthwaite's formula (Littell *et al.* 1996). Analyses were performed on natural-log transformed data. Values in bold typeface are significant with Bonferroni correction with appropriate alpha level ($\alpha = 0.004$ for height and ground-level diameter; 0.006 for 15th-year leader growth; 0.006 for foliage biomass). df and p-values for orthogonal contrasts are presented for significant treatment interactions only (cf. Table 1).

herbicide application (Table 2), or fertilizer amendment (Table 2). Annual leader growth was 16 cm for black spruce without scarification compared to 28 cm in scarified treatments; 15.5 cm without herbicide compared to 29 cm in herbicided treatments; and 19.5 cm without fertilization compared to 25 cm in fertilized treatments. The effect of *Kalmia* cover on annual leader growth (modelled for unherbicided treatments only) was not significant, either alone, or in interaction with the scarification or fertilization treatment ($p \geq 0.099$). Similarly, the significant differences in annual growth between species could not be explained by variations in *Kalmia* cover 15 years after planting ($p \geq 0.084$).

Ground-level diameter, volume index, and height:GLD ratio

Herbicide+fertilization as well as species effects on the volume index and GLD varied with the presence or absence of scarification (significant $S \times SP$ and $S \times HF$ interactions; Table 1). The herbicide effect was highly significant in both scarified and unscarified treatments ($p < 0.001$ for volume index; see Table 2 and Fig. 1D for GLD), whereas the scarification influence disappeared when herbicides were used ($p > 0.076$; see Fig. 1D for GLD). Scarification had a positive and significant effect on spruce and pine 15-year GLD and volume index ($p < 0.001$), but no effect on tamarack GLD ($p = 0.058$). As a result, the significant advantage of tamarack over jack pine in terms of GLD was eliminated in scarified treatments (Table 2; Fig. 1E). The use of herbicide significantly increased 15-year GLD and volume index of all three species, but only black spruce was positively affected by fertilization ($p < 0.001$ for volume index; see Table 2 and Fig. 1F for GLD).

The height:GLD ratio increased slightly in scarified compared to unscarified treatments (from 53 to 57; $p = 0.002$), and was reduced by 15% in herbicided treatments compared to unherbicided treatments (from 60 to 51; $p < 0.001$). Fertilization had no effect on the ratio ($p = 0.496$), whereas species were different from each other ($p < 0.001$) and were in the order spruce (45) < pine (56) < tamarack (64).

Foliar biomass

Estimated foliar biomass of individual trees increased significantly in scarified treatments (0.91 kg) compared to unscarified treatments (0.64 kg; Table 1). Species effects varied with herbicide+fertilization treatments (significant $HF \times SP$ interaction; Table 1). Estimated foliar biomass of all species increased with herbicide application. Jack pine was the only species where estimated foliar biomass was not greater in fertilized treatments compared to unfertilized treatments (Table 2, Fig. 2). Foliar biomass was in the order spruce < pine < tamarack in all herbicide+fertilization treatments, except in H1F1 where spruce = pine ($p = 0.424$, Fig. 2).

Percent *Kalmia* cover

We did not detect any significant scarification or species effects on *Kalmia* cover classes ($p > 0.132$) under individual trees. Overall, about 62% of the saplings had less than 25% *Kalmia* cover in a 0.5-m radius circular plot around their trunks. However, we observed a significant ($p < 0.001$) effect of the herbicide+fertilization treatment on sapling distribution in the various *Kalmia* cover classes. Orthogonal contrasts confirmed that the probability of trees in herbicided

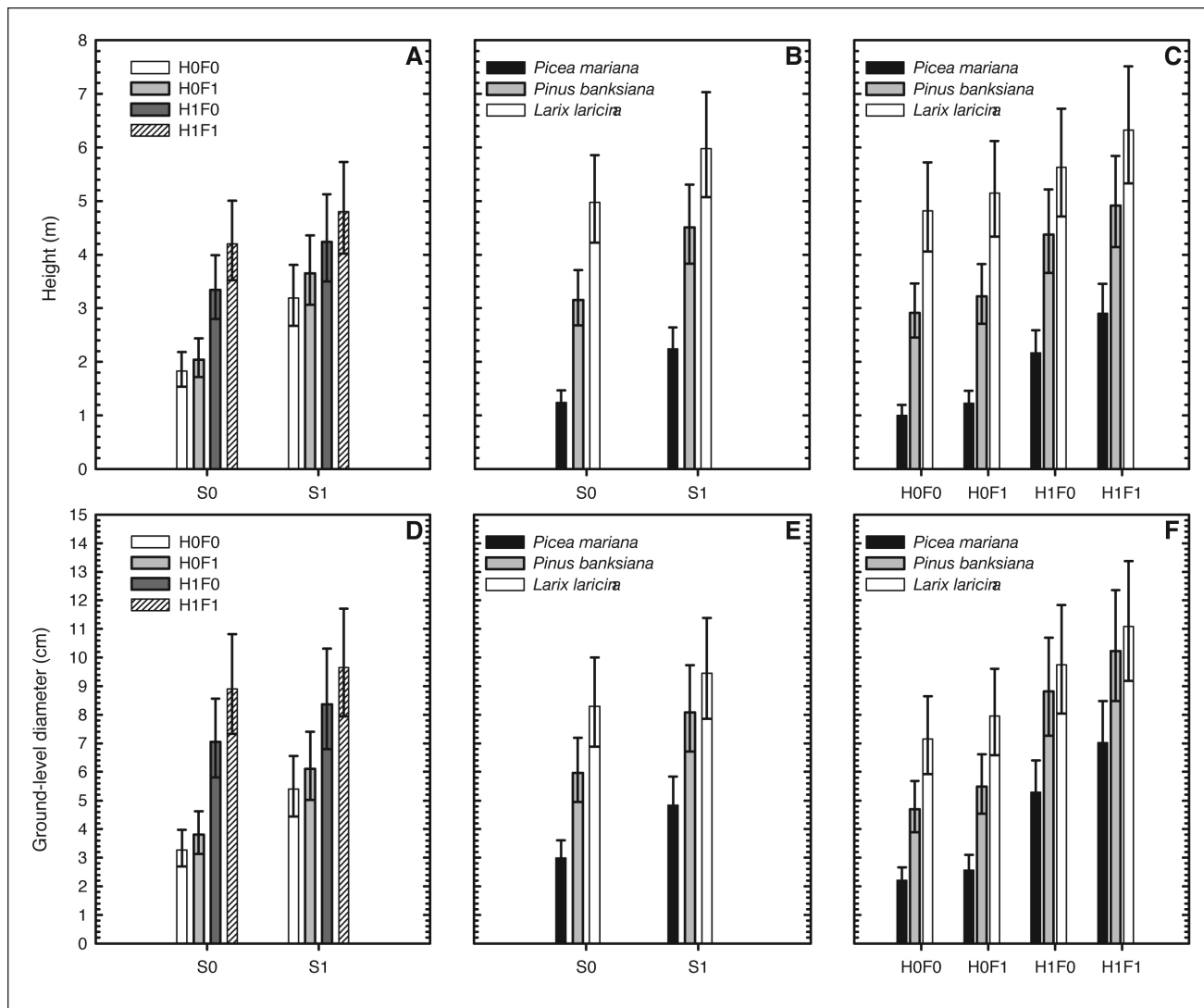


Fig. 1. Fifteen-year height (A, B, C) and ground-level diameter (D, E, F) of black spruce (*Picea mariana*), jack pine (*Pinus banksiana*), and tamarack (*Larix laricina*) saplings planted on a *Kalmia*-dominated site in central Newfoundland with various combinations of silvicultural treatments. (S0 = no scarification; S1 = with scarification; H0F0 = no herbicide, no fertilization; H0F1 = no herbicide, with fertilization; H1F0 = with herbicide, no fertilization; H1F1 = with herbicide, with fertilization.)

treatments (either fertilized or not) being associated with lower *Kalmia* cover classes was 285 times the probability of trees from unherbicide treatments being associated with lower cover classes ($p < 0.001$; Fig. 3), confirming the long-term efficacy of chemical control of *Kalmia*.

Discussion

Herbicide applications between age 1 and 4 were effective in maintaining low *Kalmia* cover in the vicinity of planted trees for more than a decade; the vast majority (98%) of the saplings in the herbicided treatments had less than 25% *Kalmia* cover in their immediate surrounding 15 growing seasons after planting. Our method of estimating *Kalmia* cover might have underestimated *Kalmia* occupancy at the plot level, however, as tree survival was likely greater in areas of low *Kalmia* cover. Nonetheless, our finding confirms—at a finer scale of resolution—that of Moroni *et al.* (2009) made at the plot level: multiple herbicide applications with an array of

chemicals can be an effective vegetation management tool for *Kalmia*. However, herbicide efficacy trials have since demonstrated the effectiveness of combining a mixture of glyphosate with a silicon-based surfactant to significantly reduce both *Kalmia* cover and stem density three and five years following a single late-August application compared to pre-treatment levels (English and Titus 2000, Titus and English 2000). Therefore, use of an array of chemicals applied more than once as at Botwood would not be necessary to obtain similar efficacy.

Given the effectiveness of the herbicide treatment in controlling *Kalmia*, it could be concluded that the significant and positive herbicide effects on tree height, GLD, volume index, and foliar biomass were the result of maintaining *Kalmia* cover below 25% over time. However, scarification also had a very significant and beneficial impact on tree dimensions and foliage: for example, the scarification effect on 15-year total height was equivalent to the herbicide effect for jack pine and

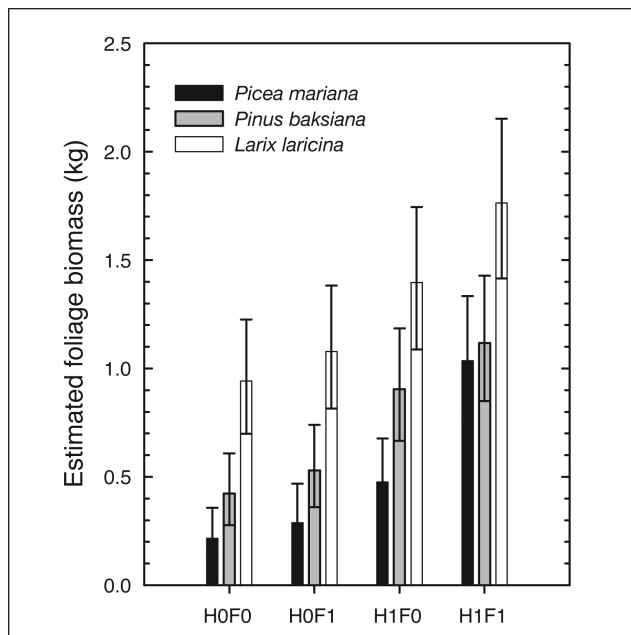


Fig. 2. Estimated foliar biomass (based on Lambert *et al.* 2005) of three conifer species 15-year after planting on a *Kalmia*-dominated site in central Newfoundland. (H0F0 = no herbicide, no fertilization; H0F1 = no herbicide, with fertilization; H1F0 = with herbicide, no fertilization; H1F1 = with herbicide, with fertilization.)

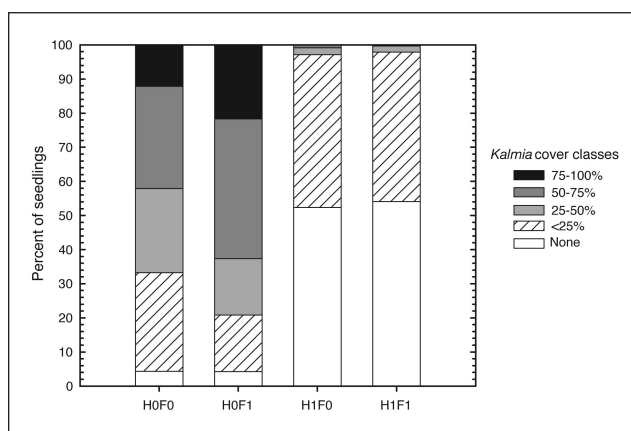


Fig. 3. Distribution of *Kalmia* cover classes in a 0.5-m radius around black spruce (*Picea mariana*), jack pine (*Pinus banksiana*), and tamarack (*Larix laricina*) saplings 15-year following planting on a *Kalmia*-dominated site in central Newfoundland. (H0F0 = no herbicide, no fertilization; H0F1 = no herbicide, with fertilization; H1F0 = with herbicide, no fertilization; H1F1 = with herbicide, with fertilization.)

tamarack, and just slightly inferior for black spruce. This positive effect of scarification was found despite the fact that it did not result in maintenance of low *Kalmia* cover over time: 37% of saplings in scarified treatments had more than 25% *Kalmia* cover around them after 15 years. Thiffault *et al.* (2005) documented that *Kalmia* ingrowth is rapid following disk trenching of ericad-dominated sites, and it is possible that the same rapid ingrowth occurred at Botwood. Thus, our results suggest that the main benefit of both scarification and herbicide treatments on tree growth at Botwood was to reduce the direct and indirect competitive pressure from *Kalmia*

during the first few years following treatment application (Thiffault *et al.* 2004b, 2005; Thiffault and Jobidon 2006), and thereby provide sufficient levels of critical environmental resources to allow seedlings to become well established. Mechanical site preparation and vegetation control can indeed significantly increase soil temperature in the rooting zone, soil water availability, and mineralization rates, especially in boreal conditions (Örlander *et al.* 1990, Prévost 1992, Wood and von Althen 1993, Smolander and Heiskanen 2007).

The difference in the height:GLD ratio between scarified and herbicided treatments, however, suggests that the treatments did not have the same level of impact on resources over time. A decreased ratio (as found in herbicided compared to control treatments) suggests that resource availability and photosynthate production were such that primary meristems expressed their full potential growth (i.e., apical growth) and that additional resources were more readily available for secondary meristems (i.e., radial growth) (Lanner 1985). On the other hand, an increase in the height:GLD ratio is an indication of unbalanced resource availability. This interpretation is compatible with the relative change in estimated foliar biomass (i.e., source of carbohydrates), which was doubled by herbicide application compared to the control treatment, and increased by 42% following scarification. Nevertheless, the height:GLD ratio remained well below critical levels for plantation stability (Jobidon 2000) in all cases. Overall, the reinvasion of treated plots by *Kalmia* had little impact on tree growth. Indeed, we did not detect any significant impact of *Kalmia* cover on 15th-year leader growth. Rather, the initial accelerated growth of the seedlings in the treated plots (either scarified or herbicided) and concomitant increase in litter production, may have initiated a positive feedback loop that increased soil fertility over time (Moroni *et al.* 2009).

Fertilization at the time of planting had limited effects on 15-year sapling dimensions. Of all three species, only black spruce was significantly influenced by this treatment, with the exception of estimated tamarack foliar biomass. In all cases, the relative gains with fertilization were much smaller than those with scarification or herbicide applications. We believe that the method of delivering fertilizer nutrients to planted seedlings is critical on ericad-dominated sites. Past work has shown that *Kalmia* and related species dominate nutrient uptake because of their large belowground biomass (Thiffault *et al.* 2004b). Thus, broadcast application of fertilizer at Botwood probably resulted in a high proportion of the nutrients being absorbed by the ericaceous shrubs. The use of spot application of slow-release fertilizers buried in the vicinity of the conifer root systems might have been a more effective way of delivering nutrients to the crop species on this site (Thiffault *et al.* 2005). However, proper mechanical site preparation so that conifer root growth is stimulated and nutrient absorption is enhanced appears to be essential for this approach to be effective (Thiffault and Jobidon 2006).

Our results are in keeping with those of previous studies regarding the relative performance of the three planted species. Tamarack is considered one of the fastest-growing native conifers in the boreal forests of North America (Johnston 1990), under both well-drained and bog conditions (Hall 1983). For example, tamarack at Botwood attained three times the height of black spruce after 15 years in scarified + fertilized treatments, a relative gain similar to that reported by

Thiffault *et al.* (2004a) after ten growing seasons on a site in northeastern Quebec with similar climate and site conditions. The superior growth of tamarack and pine compared to spruce reflects the basic differences between shade-intolerant species with high potential rates of resource capture (e.g., high photosynthetic and nutrient uptake rates) relative to shade-tolerant species adapted to low-resource environments (e.g., low photosynthetic and N uptake rates) (Reich *et al.* 1998). However, the dissimilar responses of different species to silvicultural treatments are more interesting than their relative performance *per se*. For example, growth of pine and tamarack after 15 years (i.e., 15th-year leader growth) was no longer influenced by scarification, herbicide or fertilization treatments, while black spruce leader growth was almost doubled by scarification or herbicide application, and increased by 28% with fertilization. These results differ from reports that height of seedlings of jack pine respond more than black spruce to fertilization during the early establishment phase (Thiffault *et al.* 2005), and the greater 10-year responses of tamarack and pine to silviculture than black spruce (those of Thiffault *et al.* 2004a). The reasons for these discrepancies are not obvious, and our results suggest that tamarack and pine were less affected than spruce by low resource availability and the competitive pressure of *Kalmia* found in the control treatments. However, continuous monitoring of growth over time since planting, as well as physiological and nutritional measurements of the three species, would be necessary to elucidate processes. So far, all mechanistic studies of these ecosystems have targeted *Kalmia*-black spruce interactions; the interactions between *Kalmia* and other conifer species has yet to be investigated.

Despite the lack of clear process-based explanations, our results have direct management implications for *Kalmia*-dominated sites in eastern Canada. The main objective of any silvicultural approach for managing these sites should be to create planting microsites that will remain free of *Kalmia* influences long enough for the seedlings to become well enough established that they can start influencing their own micro-environment. This, in turn, should lead to the exclusion of *Kalmia* later in the rotation through natural successional pathways. If permitted, chemical site preparation is an effective silvicultural tool for promoting planted conifer establishment, and continuous vegetation control over time should not be required. However, mechanical site preparation of sufficient intensity can lead to similar results and is thus an appropriate silvicultural alternative to herbicides. This finding is of particular and immediate relevance to the paradigm shift from traditional silviculture to more ecologically based management approaches; for example, recent papers describe the difficulty in choosing between mechanical and chemical approaches when balancing treatment effectiveness with environmental consequences or public perception of forest management (Fortier *et al.* 2005, Fortier and Messier 2006). In the boreal forest, scarification is viewed as a “near-nature” treatment because it disrupts the organic layer and exposes mineral soil, as often happens in wildfires (Fenton *et al.* 2009). The comparison of long-term effects on conifer dimensions of scarification with herbicide application thus provides forest managers with empirical evidence on which to base their decisions. Fertilization at time of planting may also confer an advantage to black spruce seedlings; however, the relative gains are marginal and cannot replace proper preparation of

the site, either chemically or mechanically. Our results also support a diversification of species choice in reforestation programs. For example, infill planting with black spruce on sites such as Botwood without prior mechanical scarification would be of little benefit because the planted trees would contribute only marginally to stand productivity (with the possible exception of seedlings planted in skid trails); however, planting tamarack with no site preparation, either mechanical or chemical, would be a viable option for attaining assured—although not optimal—reforestation or afforestation objectives on *Kalmia*-dominated sites.

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